



Impact of doxycycline discharge kits on appropriate treatment of sexually transmitted infections among patients in the ED

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ABSTRACT

Background: *Chlamydia trachomatis* is the most prevalent, reportable sexually transmitted infection (STI) in the United States. In 2021, the Centers for Disease Control and Prevention (CDC) updated treatment recommendations from a single azithromycin 1000 mg dose to doxycycline 100 mg twice daily for seven days for the treatment of chlamydia infections. In response to changes in treatment recommendations and addressing patient barriers to treatment, pharmacists at an urban, academic medical center collaborated with the state health department to create doxycycline kits dispensed upon emergency department (ED) discharge.

Objective: To evaluate if ED doxycycline discharge kits improve appropriate and timely treatment of chlamydia.

Methods: This was a single center, retrospective, chart review study of adult patients with a positive chlamydia test while seen in the ED between September 1, 2021 and September 30, 2023. Patients with a listed allergy to doxycycline and those who were incarcerated and/or pregnant were excluded. The primary outcome was appropriate, guideline recommended chlamydia treatment before and after doxycycline discharge kit implementation. Secondary outcomes included rate of ED return visits within 90 days for STI-related complaints, time to treatment in those who received a prescription for doxycycline versus doxycycline kit, and rate of doxycycline initiation via emergency medicine (EM) pharmacist culture call-back list.

Results: A total of 170 patients were enrolled in the study, 72 patients in the pre-kit group and 98 patients in the post-kit group. Significantly more patients received appropriate treatment in the post kit group (pre-kit (45.8 %) vs post-kit (69.1 %); CI 95 % 2.63 (1.4–5.0); $p = 0.002$). Time to definitive treatment was significantly shorter in the post-kit group (22.7 h pre-kit vs 1.3 h post-kit; $p < 0.001$).

Conclusions: Doxycycline discharge kits significantly increased guideline-directed treatment and decreased time-to-treatment for chlamydia in the ED population at an urban academic medical center.

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1. Introduction

Chlamydia trachomatis is the most prevalent, reportable sexually transmitted infection (STI) in the United States (US). Approximately 2 million cases were reported to the Centers for Disease Control and Prevention (CDC) in 2019 [1]. Asymptomatic infection is common in both men and women; however, medical sequelae that may result from chlamydia infection among women include pelvic inflammatory disease (PID), ectopic pregnancy, and infertility [2–4].

Historically, azithromycin 1000 mg by mouth (PO) as a single dose was the standard of care for treatment of chlamydia infections. In recent

years, several studies have demonstrated that azithromycin was inferior to other modalities of treatment for chlamydia. Treatment of rectal chlamydia with doxycycline in men who have sex with men (MSM) was superior to treatment with azithromycin for microbiological cure at 4 weeks (74 % vs 100 %; $p < 0.001$) [5]. In addition, doxycycline was significantly better for treatment of rectal chlamydia in women compared to azithromycin (79 % vs 96 %; $p < 0.01$) [6]. This is important as 80 % of women who test positive for vaginal chlamydia were found to have concomitant rectal chlamydia. It is hypothesized that there is an autoinoculation effect from the rectum to the genital area, posing a risk of adverse reproductive outcomes if left untreated [7].

As a result of this emerging data, the CDC updated their guidelines in 2021 to recommend treatment with doxycycline 100 mg PO twice a day (BID) for 7 days [2]. Patient populations with the highest rates of STIs include ages 18–26, minorities, urban area residence, and low socioeconomic status. These patient groups are high utilizers of the emergency department (ED) as a resource for primary care needs [8]. Previous

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studies have demonstrated that the majority of undertreated patient populations do not return for subsequent treatment if not empirically treated [9]. The updated treatment regimen created concern regarding patient financial and transportation barriers to prescription fulfillment as well as adherence to taking medication multiple times per day, for multiple days instead of receiving a one-time dose in the ED.

In September 2022, our ED at an urban safety-net teaching hospital that sees approximately 60,000 visits per year, implemented discharge doxycycline kits as an extension to the Emergency Medicine Pharmacist Discharge Culture Management Program. The kits are composed of a 7-day supply of doxycycline (#13 doxycycline capsules, the first dose given in the ED) for the patient to take home upon ED discharge when being empirically treated for chlamydia. Doxycycline supply is provided by the Kentucky Department for Public Health and is not associated with a charge to the patient. To our knowledge, this is the first fully pharmacist-driven service of its kind designed to overcome significant patient barriers to treatment. The purpose of this study was to evaluate if doxycycline discharge kits improve appropriate and timely treatment among patients diagnosed with chlamydia in the ED.

2. Methods

2.1. Study design, setting, and population

This was a single center, retrospective, chart review study of adult patients evaluated in the ED who tested positive for chlamydia between September 1, 2021 and September 30, 2023. Eligible subjects were 18 years or older for whom *Chlamydia trachomatis* was identified in a urine, vaginal, penile, or rectal sample via swab PCR testing and/or urine samples. Potential patients were identified based on positive results from the microbiology lab reporting feature via a third-party system, TheraDoc®. These patients were reviewed for study eligibility using the electronic medical record (EMR) by the primary author. Patients included were treated with doxycycline (kit or prescription), azithromycin, or did not receive treatment. Subjects were excluded if they were pregnant, incarcerated, and/or had a documented allergy to doxycycline. The study was reviewed by the Institutional Review Board and was determined to be exempt from informed consent based on minimal to no risk to subjects.

The included patient population was divided into two treatment groups by date: pre-kit group (September 1, 2021–August 30, 2022) and post-kit group (September 1, 2022–September 30, 2023). Each treatment group was then further divided into those who received inappropriate treatment (azithromycin or no treatment) versus appropriate treatment (doxycycline) based on guideline recommendations (Fig. 1).

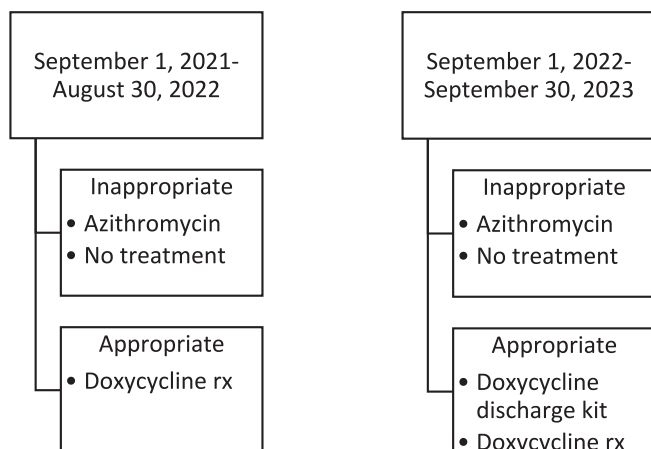


Fig. 1. Appropriate Therapy Groups by Date.

2.2. Outcome measures

The primary endpoint evaluated the number of patients who received guideline recommended chlamydia treatment before and after doxycycline kit implementation. Secondary endpoints included rate of ED return visits within 90 days of positive chlamydia test for STI related complaints, time to doxycycline treatment in those who received a discharge kit versus those who received a prescription for doxycycline, and the rate of initiation of doxycycline therapy via the pharmacist-led culture callback list before and after kit implementation.

2.3. Statistical analysis

A sample size of 62 patients in each arm was calculated to detect an anticipated difference of 20 % between groups for the primary outcome (70 % appropriate treatment pre-kit versus 90 % appropriate treatment post-kit); α value of 0.05, a β value of 0.2 (80 % power). Data was analyzed in SPSS by IBM. Chi-square test or Mann-Whitney *U* test were used for comparison between treatment groups as appropriate; discrete data are reported as frequencies and percentages, and continuous and ordinal data are expressed as mean (standard deviation [SD]) or median (interquartile range [IQR]) as appropriate.

3. Results

3.1. Patients and demographics

From September 1, 2021 to September 30, 2023, a total of 268 patients were screened, of which 170 were included: 72 in the pre-kit group and 98 in the post-kit group. The most common reasons for exclusion were pregnancy and age less than 18 years old (Fig. 2). Overall, the groups were similar regarding baseline characteristics (Table 1). Both populations predominately consisted of young, African American females. The groups were similar in the number of patients who presented with symptoms of infection such as painful urination (34.7 % pre-kit and 35.1 % post-kit) and vaginal/penile discharge (51.4 % pre-kit and 52.6 % post-kit). Notably, the post-kit group had a greater number of patients who received initial treatment in the ED (33.3 % vs 43.3 % pre-kit and post-kit, respectively). Of the 24 patients who received initial treatment in the ED in the pre-kit group, 16 received

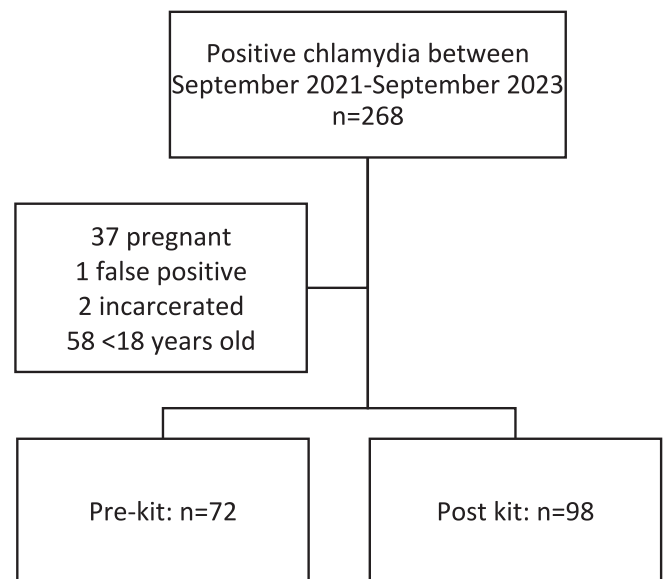


Fig. 2. Study Flowchart.

Table 1
Baseline Characteristics.

	Pre-Kit (n = 72)	Post-Kit (n = 98)
Age (years), mean [SD]	26 [7.5]	28 [11.2]
Female, n (%)	39 (54.2)	45 (46.3)
African American, n (%)	53 (73.6)	74 (76.3)
Weight (kg), mean [SD]	78 [25.9]	78 [19]
History of HIV/AIDS, n (%)	3 (4.2)	1 (1)
Painful Urination, n (%)	25 (34.7)	34 (35.1)
Vaginal/Penile Discharge, n (%)	37 (51.4)	51 (52.6)
Miscarriage, n (%)	3 (4.2)	0 (0)
Received Initial Treatment in ED, n (%)	24 (33.3)	42 (43.3)
Concomitant STI, n (%)	42 (58.3)	53 (54.1)

doxycycline and 8 received azithromycin. All 42 patients in the post-kit groups received doxycycline as their initial treatment in the ED.

3.2. Outcomes

There was a significant difference for the primary outcome of those who received appropriate treatment between groups (45.8 % pre-kit vs 69.1 % post-kit; CI 95 % 2.63 (1.4–5.0); $p = 0.002$) (Table 2). There was no difference in 90-day ED return visits for STI-related complications (9.7 % pre-kit vs 13.4 % post-kit), time to initial treatment in the ED (0.6 h [0.2–2.3 h] pre-kit vs 0.7 h [0.3–1.7] post-kit), and doxycycline prescription via culture callback program (15.3 % pre-kit vs 11.3 % post-kit). However, time to definitive treatment with doxycycline kit or prescription was significantly shorter in the post-kit group (pre-kit 22.7 h [15.4–55.1] vs post-kit 1.3 h [0.5–31.4]; $p < 0.001$) (Table 2).

4. Discussion

To our knowledge, there are no other studies designed to evaluate similar initiatives, specifically in the ED. In this study, patients with chlamydia received adequate treatment more often when discharged with a doxycycline kit. These patients also had a quicker time to definitive treatment when receiving a discharge kit, allowing for shorter time in the ED and potential for quicker cure. Three patients in the pre-kit group experienced a miscarriage upon evaluation at their initial ED visit while no patients in the post-kit group experienced a miscarriage. Although this study was not powered for this outcome, this is a notable finding as miscarriage has been found to be a medical sequela from chlamydia infection. [3,10]

The present study is not without limitations, primarily related to retrospective design. Specifically, the EMR insurance fill history only has access to those patients with paid insurance claims, meaning those who may have paid for their doxycycline prescription with cash or discount cards were counted as not receiving adequate treatment. We were unable to confirm appropriate treatment in 34 % of patients in the pre-kit group and 14 % of the post-kit group due to lack of fill history. Additionally, the insurance fill history does not have a time listed

for prescription pick-up. Due to this, the assumed time for prescription pick-up for all patients was 18:00. This assumption could potentially have led to a skewing in the time to definitive treatment. Furthermore, a limitation exists due to the inability to see if patients returned to another ED outside of our health-system. It is important to note that variations in prescribing practices and non-adherence to treatment guidelines could be due to provider preferences and adoption of newer guideline recommendations overtime. Finally, the 90-day ED return outcome may be representative of re-infection versus actual microbiological failure.

Despite the listed limitations, this study has several notable strengths. This study addresses a unique treatment option for a major public health concern and to our knowledge, our ED is among the first health systems to provide this service to the community. The use of doxycycline discharge kits removed several significant barriers to treatment by decreasing the financial and logistical burden of outpatient pharmacy transportation in our patient population.

The success of this program opens the door for further initiatives with doxycycline discharge kits including expanding the program into further ED settings across the nation. With the support of state health departments, accessible, preferred treatment is a possibility for all. Furthermore, expedited partner therapy (EPT) is an area that the initiative can progress to once well established.

5. Conclusion

In conclusion, this study demonstrated that pharmacy-driven doxycycline discharge kits significantly increased guideline-directed treatment and decreased time to treatment for chlamydia infections in the ED population at an urban academic medical center. Overall, this initiative overcame barriers to treatment for a significant public health issue, supporting the need for expansion to other emergency departments across the country.

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CRedit authorship contribution statement

Carly Loudermilk: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Pauline Thiemann:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Joshua Senn:** Writing – review & editing, Supervision, Conceptualization. **Jacob Shreffler:** Writing – review & editing, Validation, Software, Formal analysis.

Declaration of competing interest

None.

Table 2
Outcomes.

	Pre-Kit (n = 72)	Post-Kit (n = 98)	Odds Ratio (95 % CI)	P-value
Primary Outcome				
Appropriate Treatment, n (%)	33 (45.8)	67 (69.1)	2.63 (1.4–5.0)	0.002
Secondary Outcomes				
90-Day Return, n (%)	7 (9.7)	13 (13.4)	–	–
Time to Initial Treatment (hours), median (IQR)	0.6 (0.2–2.3)	0.7 (0.3–1.7)	–	–
Time to Doxycycline Kit/Rx (hours), median (IQR)	22.7 (15.4–55.1)	1.3 (0.5–31.4)	–	–
Doxycycline Rx via Culture Call Back, n (%)	11 (15.3)	11 (11.3)	–	< 0.001

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